NATIONAL GRID ANALYTICS | SPECIAL FESTIVE STUDY

The Diwali Electricity Paradox

Why does India's national power consumption plummet by 15% to 19% during the Festival of Lights?

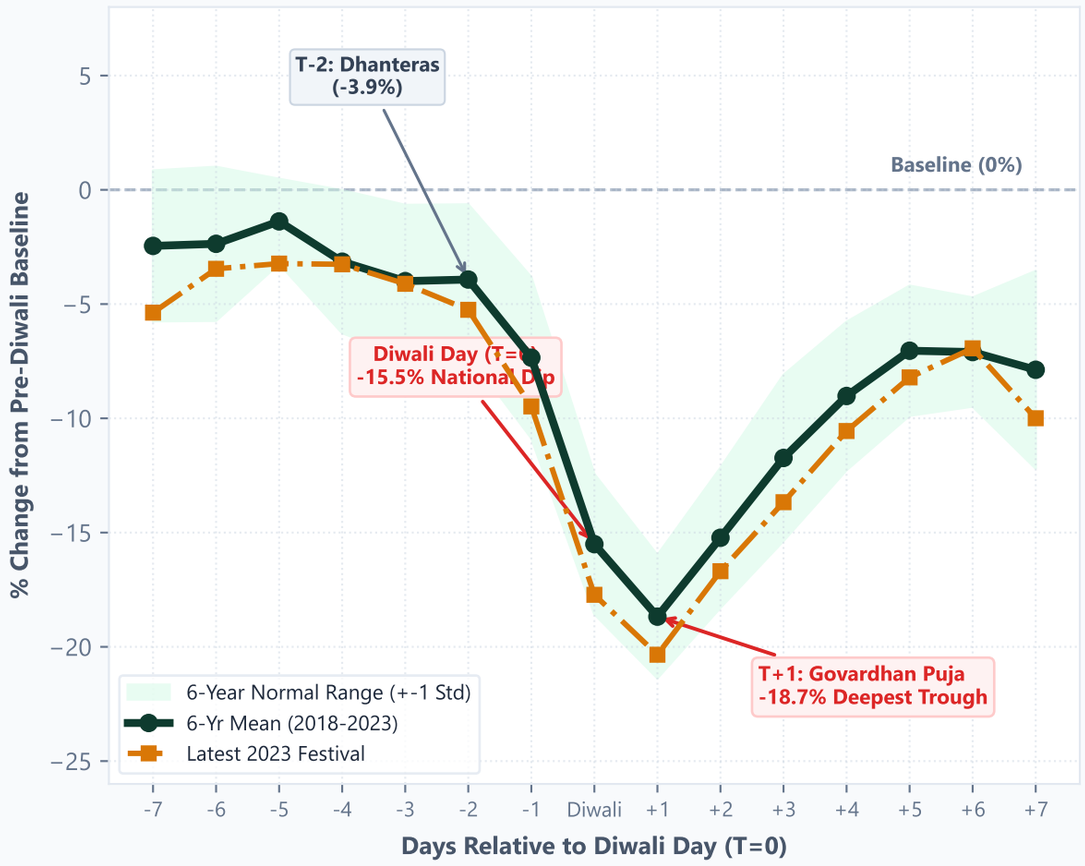
-15.5%
Diwali Day Deficit
11-yr avg drop on festival day

-18.7%
Trough at T+1
Govardhan Puja / New Year peak dip

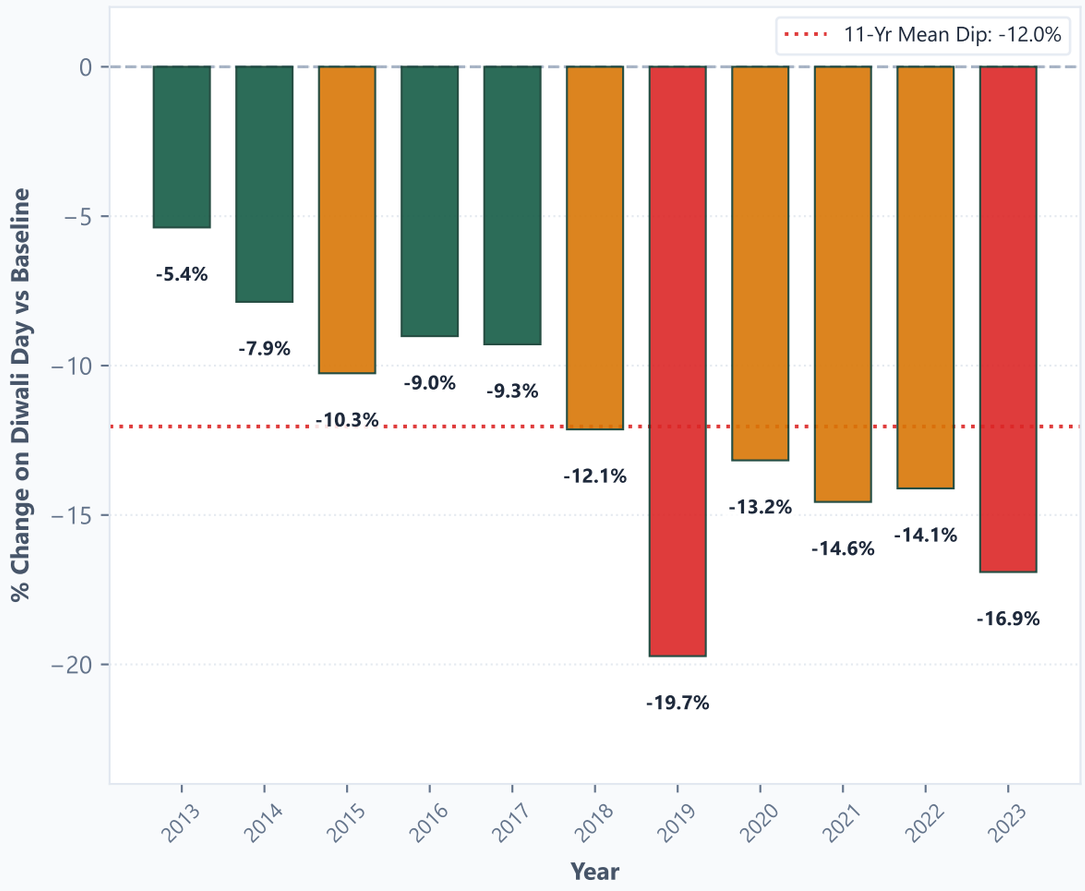
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Decadal Consistency
Every year 2013-2023 saw a drop

-47.5%
Peak State Dip
Daman & Diu (TN -31.8%, GJ -28.8%)

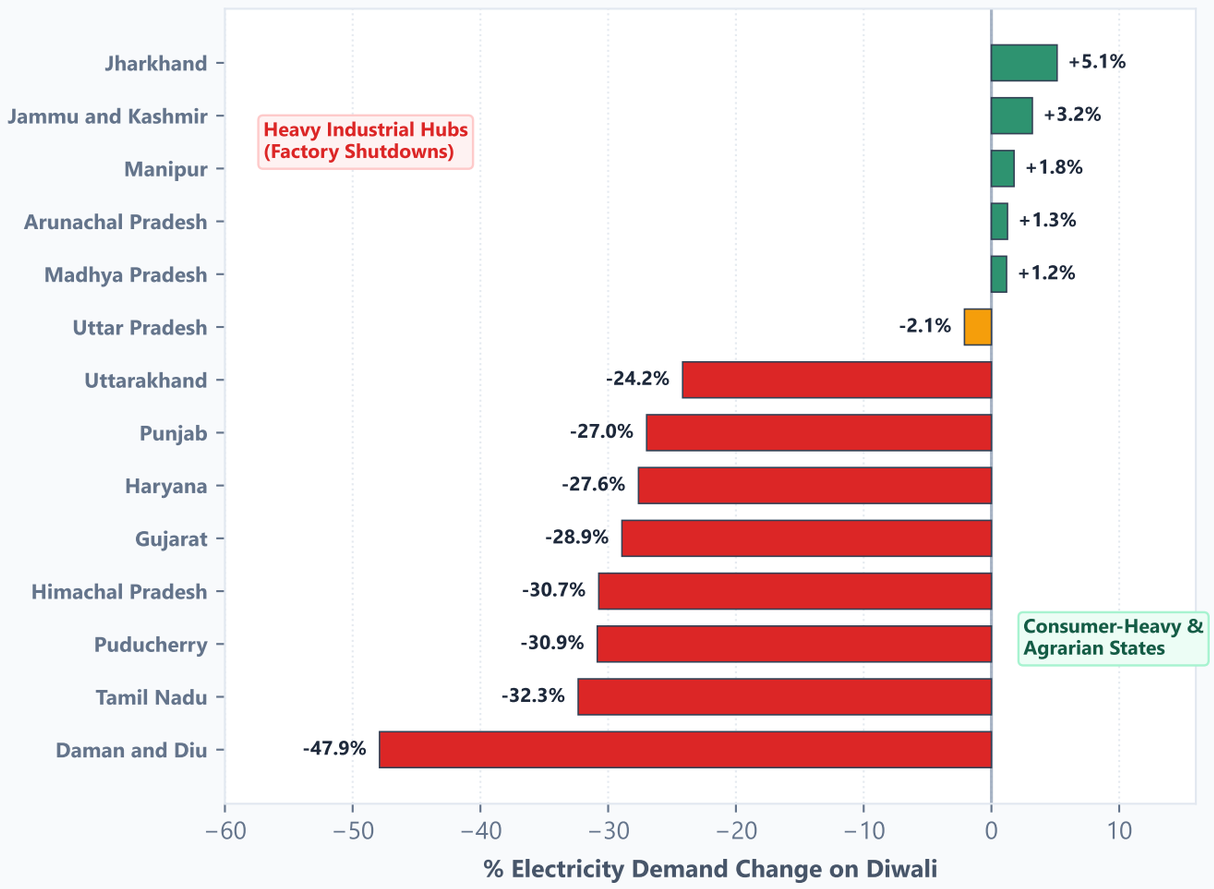
A. The 15-Day Demand Trajectory (T-7 to T+7)



B. 11-Year Historical Consistency (2013-2023)



C. State Divergence: Industrial Halts vs Consumer Resilience



D. Grid Physics: The 3 Drivers of the Paradox

1. The Commercial & Industrial (C&I) Shutdown Effect

Commercial and industrial customers account for ~49% of all electricity consumed in India. Continuous-process plants (textiles, chemicals, auto manufacturing, steel rolling) in Gujarat, Tamil Nadu, and Maharashtra initiate planned annual maintenance halts during Dhanteras through Bhai Dooj. The idling of multi-megawatt industrial motors creates a deficit that domestic loads cannot replace.

2. The Modern LED Efficiency Revolution

While public intuition assumes decorative lighting creates massive power surges, modern LED serial strings draw only 3W to 10W per 10-meter set (compared to legacy 60W-100W incandescent fixtures). Even if 100 million households power 3 LED strings simultaneously, the total peak lighting load is ~1,500 MW for 4 hours (~6 MU), which is barely 1% of the ~686 MU daily industrial drop.

3. Agricultural Pumping Holidays

Agricultural irrigation pumpsets represent ~18% to 22% of total load in major states (Punjab, Haryana, Rajasthan, Madhya Pradesh). Tubewells and feeder pumps are idled by farmers during the festival festivities, amplifying the grid-wide demand suppression.

EXECUTIVE GRID SUMMARY & UTILITY IMPLICATIONS

For power system operators (Grid-India, SLDCs), Diwali is not a generation-scarcity challenge, but a high-voltage regulation and over-frequency challenge. With ~15% to 19% of national demand suddenly vanishing, transmission grids experience severe Ferranti effects (voltage spikes from lightly loaded lines). Grid managers actively back down thermal base generation, ramp down hydro, and switch in 765 kV bus reactors to maintain 50 Hz grid stability.

Data: Grid-India / NLDC Open Data (3,955 Days, 132,600+ State Records) | Analysis: Analysis Hub | Author: Yash Vasudeva